

AARIZ ZAFAR

Machine Learning Engineer | Cloud

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PROFESSIONAL SUMMARY

Computer Science Engineering graduate with 1+ year of industry experience in machine learning, data science, and cloud AI. Built ML solutions for predictive maintenance, synthetic data generation, industrial time-series analytic, RAG systems, and computer vision-based safety monitoring.

TECHNICAL SKILLS

- Machine Learning & Data Science: supervised learning, anomaly detection, time-series analysis, feature engineering, model evaluation, statistical modelling
- AI & Computer Vision: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, object detection, PPE compliance monitoring
- Generative AI & NLP: LangChain, LangGraph, RAG, vector databases
- Data Engineering & Cloud: PySpark, Apache Spark, Kafka, Airflow, Hadoop, Azure Data Factory, GCP Dataproc
- DevOps & Deployment: Docker, Jenkins, GitHub Actions, Pytest
- Programming & Databases: Python, C++, Go, Java, SQL, PostgreSQL, MySQL, MongoDB
- Domain Expertise: Predictive Maintenance, Fault Detection & Diagnosis, Computer Vision for PPE Compliance Monitoring

EXPERIENCE

Trinity Mobility - AI Engineer, Bangalore, India Present

Mar 2026 -

- Simulated 10+ degradation fault scenarios with Energy Plus and Modelica to create synthetic sensor datasets for HVAC where real-world fault data was limited.
- Utilized pre-trained time-series forecasting models (TimesFM, Chronos) for predictive maintenance and equipment health monitoring, improving forecasting accuracy by 15%.

LivNSense - Software Engineer, Bangalore, India

Dec 2024 - Mar 2026

- Developed an end-to-end ML pipeline for asphalt plant **optimization** using EDA, statistical modelling, and anomaly detection, improving operational efficiency by 10%.
- Designed and deployed a domain-specific RAG solution using LangChain, and vector embedding, reducing engineering knowledge retrieval effort by 30%.
- Built a C++ image pre-processing pipeline for industrial PPE detection, improving object detection accuracy from 75% to 87% for real-time monitoring of 100+ workers.

NETTECH - Software Engineering Intern, Bangalore, India

Oct 2023 - Dec 2023

- Developed a RAG system using LangChain and SBERT embeddings to support natural language querying of network device documentation, reducing lookup effort by 20%.

PROJECTS

NanoGPT

- Built and deployed a NanoGPT project to demonstrate **transformer**-based language modelling concepts on **Azure**.
- Applied tokenization, model inference workflow design, and cloud deployment practices to make the project easy to inspect and share.

BPE Tokenizer

- Built a Byte Pair Encoding tokenizer demonstrating tokenization and embedding preparation techniques used in large-scale transformer models, achieving up to 25% text compression.
- Containerized and deployed the platform with Docker, Azure Static Web Apps, Container Apps, Container registry.

EDUCATION

Vellore Institute of Technology AP

2021- 2025

B.Tech, Computer Science and Engineering | CGPA: 8.66/10.0

CERTIFICATIONS

- Microsoft Certified: Azure Security Engineer Associate (AZ-500) | Azure Administrator Associate (AZ-104) | Azure Fundamentals (AZ-900)
- Machine Learning - Boston Training Academy | AI & ML Program - Google Developer / SmartInternz

AWARDS & INDUSTRY ENGAGEMENT

- Received Star Performer Award and Certificate of Appreciation at LivNSense for contributions to AI-based industrial optimization projects.
- Represented LivNSense at Automation Expo 2025 at the Bombay Exhibition Centre, Goregaon, demonstrating AI-driven industrial automation solutions to industry partners.